

Load TSV file and print dataset statistics using R

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Loads a TSV (tab-separated) file using R's `read.delim` and prints rows, columns, total cells, file size, and object memory usage.

Language

rscript

Category

data-processing

Command

```
Rscript -e "args<-commandArgs(trailingOnly=TRUE);if(length(
  args)<1)stop('Need filename');df<-read.delim(args[1]);cat
  ('Rows:',nrow(df),'Cols:',ncol(df),'Cells:',nrow(df)*ncol
  (df),'File size:',file.size(args[1]),'Object size:',
  format(object.size(df),units='auto'),'\\n')"
```

Explanation

The command uses Rscript to execute R code that reads a TSV file specified as argument, computes basic statistics (rows, columns, cells), gets file size via `file.size()`, and reports object memory usage via `object.size()`. `read.delim` uses tab as separator by default.

Tags

rscript, tsv, statistics, data-analysis, memory

Dependencies

r-base-core

Arguments

1. **TSV_FILE** (Required): TSV (tab-separated) file to analyze

Examples

1. `Rscript -e "args<-commandArgs(trailingOnly=TRUE);if(length(args)<1)stop('Need filename');df<-read.delim(args[1]);cat('Rows:',nrow(df),'Cols:',ncol(df),'Cells:',nrow(df)*ncol(df),'File size:',file.size(args[1]),'Object size:',format(object.size(df),units='auto'),'\\{\\}n')"`
data.tsv - Analyze data.tsv
2. `Rscript -e "args<-commandArgs(trailingOnly=TRUE);if(length(args)<1)stop('Need filename');df<-read.delim(args[1]);cat('Rows:',nrow(df),'Cols:',ncol(df),'Cells:',nrow(df)*ncol(df),'File size:',file.size(args[1]),'Object size:',format(object.size(df),units='auto'),'\\{\\}n')"`
large_dataset.tsv 2>/dev/null - Analyze suppressing warnings

Output

```
Rows: 100 Cols: 10 Cells: 1000 File size: 12345 Object size:
78.2 Kb
```

Notes

- Uses `read.delim` with default settings (`header=TRUE`, `sep='\\{\\}t'`, `stringsAsFactors=FALSE` in R_i=4.0)
- File size reported in bytes
- Object size includes R overhead
- For large files, consider adding options(`stringsAsFactors=FALSE`)

Warnings

- May load entire file into memory
- Very large TSV files could exhaust memory

See Also

- `csv-stats`

Status

draft

Safety

safe

Shell

posix

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

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